

APPLIED INORGANIC CHEMISTRY

EXTRACTION OF METALS

Metals are usually extracted from their ores.

An **ore** is a naturally occurring impure chemical substance from which an element such as a metal is obtained (extracted).

Generally the stages may be employed in obtaining the metal from the ore include:

Stage 1: Concentration of the ore.

Stage 2: Roasting of the concentrated ore.

Stage 3: Extraction of the metal from the ore.

Stage 4: Purification of the extracted metal.

Concentration (Dressing or Benefaction) of the ore

This is the process by which the ore from which the metal is obtained is made richer in the metal ore rather than the impurities by removing the large bulk of impurities known as the gangue.

Ores obtained from the ground have a large bulk of impurities unlike those from water bodies.

There are numerous methods of concentration and the methods are chosen based on the properties of the ore.

The advantage is that concentration of the ore makes the process of extraction economical.

(a) Froth floatation.

This is a physical method. Most metal ores with low densities especially sulphide ores like zinc blende, galena, etc are concentrated by the process of **froth floatation (selective wetting)**.

The process uses the difference in the wetting characteristics of minerals and gangue. Minerals are generally wetted by oil and gangue is wetted by water.

In froth floatation, the ore is finely crushed to increase the surface area. The crushed ore is then mixed with water and treated with frothing agents or froth stabilizers such as pine oil (or soaps or cresols) together with collectors such as fatty acids. The mixture is then agitated.

The oil wets the ore particles while water wets the unwanted materials called gangue.

Compressed (a blast of) air is then introduced into the base of the froth floatation cell containing the mixture. Small air particles attach themselves to the oil particles and carry the wetted and less dense ore particles to the surface to form a froth. The froth spills over (floats on top) and the mineral particles in it are skimmed off.

The ore is broken down by adding concentrated sulphuric acid, filtered and dried.

Note that:

- Fatty acids increase the non-wettability of mineral particles by forming a film on the mineral particles which makes them water repellent.
- Pine oil stabilizes the foam that rises to the top, to make the separation easier.
- Depressants may be other suitable additives: Sometimes two similar minerals are both found in the ore and both react to collectors similarly. This will mean that both will rise to the top of the foam. To separate such similar compounds we use depressants.
e.g. in an ore containing both ZnS and PbS (both sulphides), sodium or potassium cyanide, NaCN is added as a depressant. It reacts with ZnS not allowing it to froth and only PbS comes up in the foam.
- Froth floatation has the advantage over gravity separation because it can collect even the extremely fine particles of minerals.

(b) Hydraulic washing.

This is a physical method based on the difference in weight between the gangue and the metal particles. The principle (physical property) of **gravity separation** is used to separate the two.

Generally, the metal particles of the ore are heavier than the impurities.

In hydraulic washing, the ore is finely crushed to increase the surface area. The crushed ore is mixed with an upward stream of running water. The lighter particles of the gangue wash away in the running water. The heavier metal particles settle down and can be separated easily.

This method is used for ores that contain lead and tin since these are relatively heavier.

(c) Magnetic separation.

This is a physical method and the property of magnetic properties of metals is used in the separation. i.e. in this method of magnetic separation, the magnetic properties of certain metals are used to separate them from the non-magnetic gangue.

In magnetic separation, the ore is finely crushed to increase the surface area. The crushed ore is passed on a conveyor belt passing over a magnetic roller. The magnetic ore particles are attracted and remain on the belt while the non-magnetic gangue falls off the belt.

Alternatively, the reverse may be the case. i.e. the gangue may be the magnetic material.

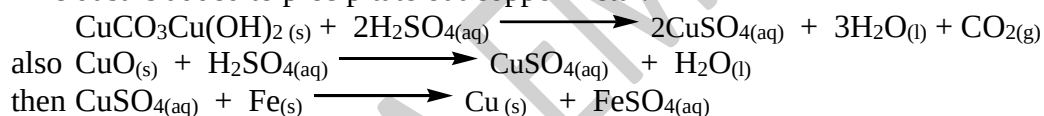
Iron ores and manganese ores are the main ones for which magnetic separation is used.

Magnetite and chromite ores use this method on a large scale to remove the gangue.

(d) Leaching.

This is a chemical method which involves extraction of a metal as a soluble salt by dissolving the powdered ore in a suitable chemical solvent (aqueous solution). The chemical solution dissolves the metal in the ore leaving behind the insoluble impurities which are filtered out. The purified ore or metal in solution can be separated from the soluble impurities by precipitation.

i) In obtaining copper from its ores malachite, $\text{CuCO}_3\text{Cu(OH)}_2$ and the copper(II) oxide ore, CuO . The ore is dissolved with warm dilute sulphuric acid to form copper(II) sulphate solution. Iron or zinc dust is added to precipitate out copper metal.



ii) In obtaining manganese from its ore pyrolusite, impure manganese(IV) oxide. The ore is leached using warm dilute sulphuric acid.

iii) The Bayer process in purification of the aluminium ore, bauxite. The ore is dissolved in warm concentrated sodium hydroxide, seeded to form aluminium hydroxide which is then heated to obtain the purified ore.

(e) Calcination.

This is mainly a chemical method of purification. The word calcination is derived from the Latin word “calcinare” which translates as “to burn lime”.

Calcination is the process in which the **ore** of the metal is heated to high temperature in the absence or limited supply of air or oxygen. During calcination, solids are heated in a controlled environment to mainly remove volatile substances, water or oxidize the substance. e.g.

i) Calcination of limestone involving decomposition of the carbonate ore and removal of carbon dioxide to form quick lime.

ii) Calcination of gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) involving the removal of water of crystallization in the form of water vapour giving different products such as Plaster of Paris or dental plaster ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$), dental stone, etc.

iii) Decomposition of volatile components from raw petroleum coke.

iv) Bauxite, $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ is calcinated and water vapour is removed to form anhydrous alumina (aluminium oxide).

v) Anhydrous iron(III) oxide is formed during calcination of hydrated iron ores to remove water of crystallisation.

Roasting of the concentrated ore

Roasting is the process in which the **ore** is heated to high temperatures in the presence of excess supply of air or oxygen.

This stage is involved for ores that don't exist as oxides. For example metals with medium negative electrode potentials have a tendency to form covalent bonds and occur mainly as sulphides, carbonates.

Roasting of the concentrated ore is the process of heating mineral concentrated ores containing sulphides or carbonates to remove impurities as gases and forming metal oxides.

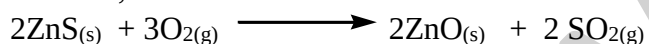
During roasting, the concentrated ore is finely ground into small pieces and then suspended in a rising stream of air.

It is then heated and either the sulphur from the metal sulphide is removed as sulphur dioxide gas or the carbon from the metal carbonate is removed as carbon dioxide gas.

The ore is converted into the metal oxide which can easily be reduced as compared to the either sulphide or carbonate.

For example, the two ores of zinc i.e. Zinc blende, ZnS and Calamine, ZnCO_3 .

For Zinc blende, ZnS is roasted in air to form Zinc oxide.

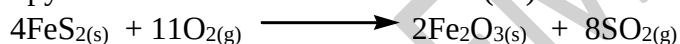


Similarly for calamine, ZnCO_3 is roasted in air to form Zinc oxide.



Another example are ores of Iron i.e. siderite, FeCO_3 and iron pyrites, FeS_2 .

For Iron pyrites is roasted in air to form Iron(III) oxide.



Similarly for Siderite, FeCO_3 is roasted in air to form Iron(III) oxide.



When an ore is roasted in air and the compacted mixed oxide called triiron tetraoxide, Fe_3O_4 is the main product, this process is known as *sintering*.

Hence the sintered ore is in form of triiron tetraoxide.

Once the oxides are formed, they are easier to deal with (reduce) in the blast furnace than the sulphides or carbonates hence can easily be reduced using carbon or carbon monoxide.

Difference between Calcination and Roasting include:

Calcination	Roasting
Calcination is the process in which the ore of the metal is heated to high temperature in the absence or limited supply of air or oxygen.	Roasting is the process in which the ore is heated to high temperatures in the presence of excess supply of air or oxygen.
Calcination consists of thermal decomposition of mostly calcium ores.	Roasting is mostly done for sulphide ores.
Carbon dioxide is given out as a chemical by-product.	During roasting, large quantities of toxic, metallic and acid impurities are driven out.

Calcination is used for removal of moisture from the ore.

Roasting is not used for the removal of moisture.

Extraction of the metal from the ore

The various methods employed depend on the reactivity of the metals. Those which are highly reactive (electropositive) such as sodium and aluminium, are obtained from their ore by **electrolysis**. However those which are less reactive such as zinc and iron are mainly obtained by **reduction** method using carbon (coke) since it is less expensive for them to run. The various methods employed include the following:

(a) Electrical Method (Electrolysis):

It is used to extract metals which are very reactive and highly electropositive. The metals are extracted by electrolysis from their fused salts or oxides.

(b) Reduction Method:

This method is used to extract metals with a high affinity for oxygen. These elements occur as oxides or silicates and the reduction of the metal oxides is carried out using carbon. Examples of such metals include iron and zinc.

Smelting is the process of heating and melting an ore to obtain a molten metal from it using a reducing agent like carbon or carbon monoxide.

It is at this stage where *most of the impurities such as silica and aluminium oxide are removed as slag* (calcium silicate or calcium aluminate).

During smelting, fluxes such as calcium carbonate (that decomposes to calcium oxide) may be added so as to help in fusing with the impurities which are removed as a slag. The fluid slag floats on the surface of the liquid metal and is then tapped off.

Refer to extraction of Copper and Iron.

(c) Displacement Method:

This method is used mainly to extract noble metals unreactive with positive electrode potentials. They occur as easily reducible compounds and are extracted by displacement using Hydrogen over their heated oxides. e.g. Copper.

Purification of the extracted metal

This is achieved generally by using the; electrolysis process, fractional distillation process, bessemer process, open hearth process.

EXTRACTION OF SODIUM

Sodium is extracted in one stage involving **electrolysis of fused or molten sodium chloride** using Down's cell (*commonly referred to as the Down's process*) as shown in the diagram below.

Fused or molten sodium chloride is mixed together with calcium chloride and then fed into the Down's cell. Electrolysis is carried out using the iron cathode and a carbon (graphite) anode.

At the anode; $2\text{Cl}^-_{(l)} - 2e^- \longrightarrow \text{Cl}_{2(g)}$ or $2\text{Cl}^-_{(l)} \longrightarrow \text{Cl}_{2(g)} + 2e^-$

At the cathode; $\text{Na}^+_{(l)} + e^- \longrightarrow \text{Na}_{(l)}$

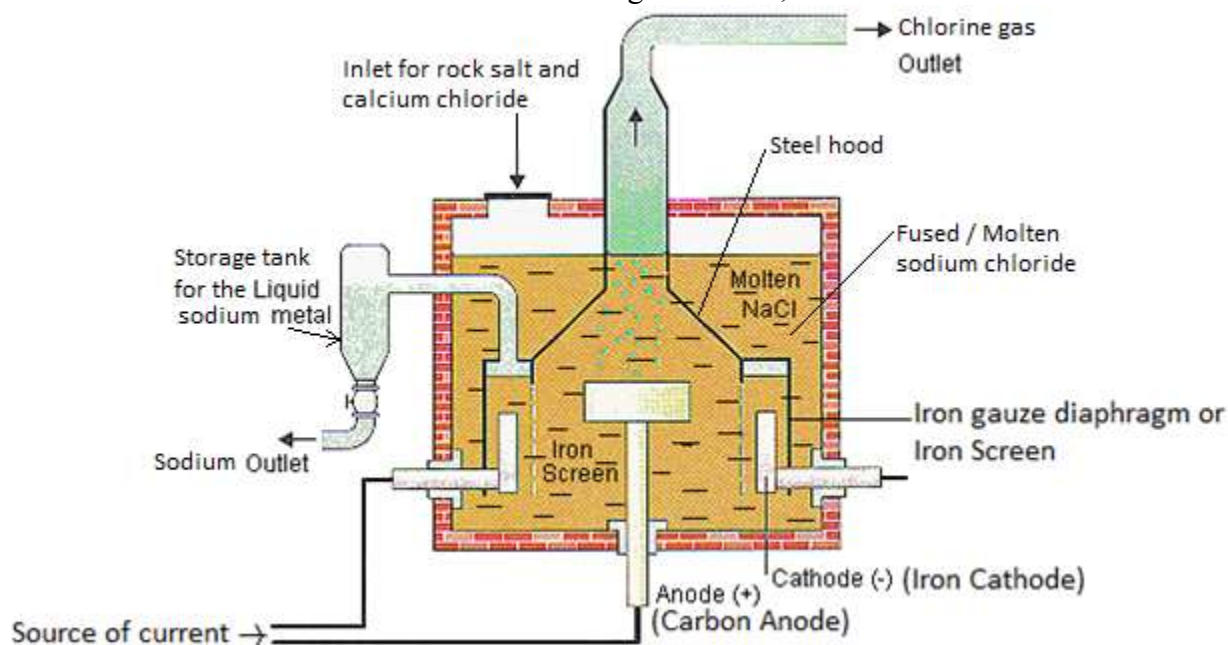
Chlorine gas is obtained at the anode while molten sodium at the cathode.

Calcium chloride is added in order to lower the melting point of sodium chloride from 1074K (i.e about 800°C) to 870K (i.e. 600°C).

Some calcium is also discharged but it does not mix with sodium since it is denser and they can easily be separated out. Calcium chloride also has an advantage of increasing the chloride ions in the cell.

Chlorine gas is another product which is also obtained during the manufacture of sodium. Thus this process can be used in the manufacture of chlorine gas.

The Down's cell is illustrated as shown in the diagram below;



Uses of Sodium

- (a) It is used in the manufacture of sodium hydroxide (NaOH).
- (b) It is used in the manufacture of sodium peroxide (Na₂O₂).
- (c) It is used in photo electric cells.
- (d) It is used in the manufacture of sodium cyanide (NaCN) which is also used in the extraction of gold.
- (e) It is used in the manufacture of sodamide (NaNH₂).
- (f) Its alloy is used in the manufacture of anti-knock additives for petrol e.g. tetraethyl lead.
- (g) Sodium is used for the withdrawal of heat from some nuclear reactors. It is therefore used as a coolant.
- (h) Sodium is used as a reducing agent in the laboratory.

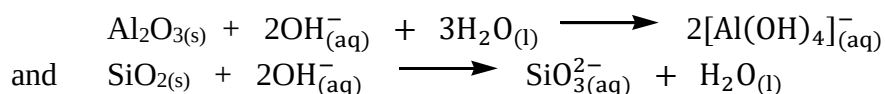
EXTRACTION OF ALUMINIUM

Aluminium is obtained by the electrolysis of molten aluminium oxide. The oxide is obtained from the ore called bauxite which is impure aluminium oxide with formula Al₂O₃.2H₂O. It contains silicon(IV) oxide and iron(III) oxide as major impurities. Stages involved are:

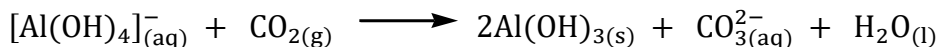
Stage 1. Concentration of the ore by leaching: In this method, a suitable solvent is used to dissolve the substance and the precipitate out the pure substance from solution.

The ore is first roasted to convert all the iron(II) present to iron(III) oxide.

It is then heated with concentrated sodium hydroxide. The aluminium oxide and silicon dioxide (silica) dissolve to form a solution but the iron(III) oxide and other basic materials remain undissolved. Thus



The basic insoluble materials are filtered off. The filtrate is then treated with carbon dioxide or seeded with aluminium hydroxide to precipitate aluminium hydroxide. Silica remains dissolved (in solution).

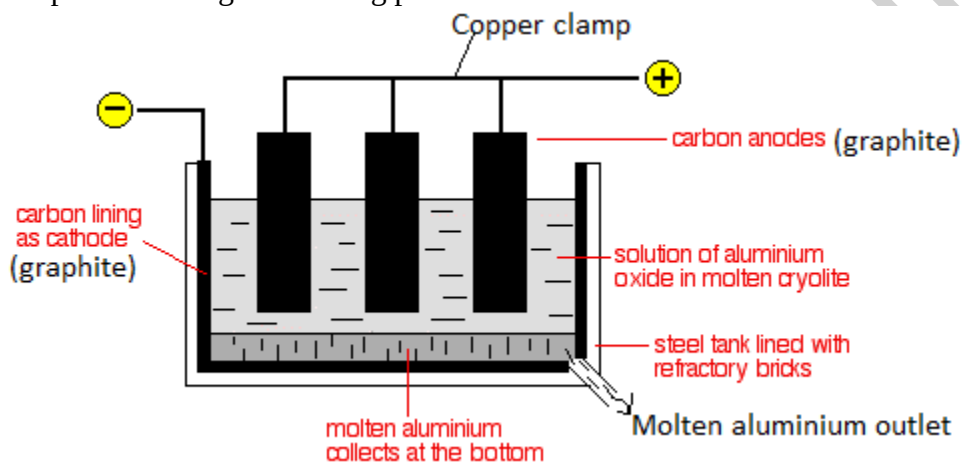


or $[\text{Al}(\text{OH})_4]^-_{(\text{aq})} \longrightarrow \text{Al}(\text{OH})_{3(\text{s})} + \text{OH}^-_{(\text{aq})}$ i.e. seeding where aluminium hydroxide induces hydrolysis of the tetrahydroxoaluminate ion to precipitate aluminium hydroxide.

The aluminium hydroxide is then heated to obtain the concentrated aluminium oxide.



Stage 2. Extraction by electrolysis of molten aluminium oxide: The oxide is then dissolved in molten cryolite (Na_3AlF_6) and electrolysed at about 900°C using graphite (carbon) electrodes. Cryolite helps in lowering the melting point of aluminium oxide and also acts as an electrolyte.



At the cathode: Aluminium is discharged as $\text{Al}^{3+}_{(\text{l})} + 3\text{e}^- \longrightarrow \text{Al}_{(\text{l})}$

The molten aluminium collects at the bottom of the cell and is tapped off periodically.

At the anode: Oxygen is given off as $2\text{O}^{2-}_{(\text{l})} \longrightarrow \text{O}_{2(\text{g})} + 2\text{e}^-$

The oxygen reacts with the carbon (graphite) of the anode to form oxides of carbon at the high temperatures involved. As a result the anodes are gradually burnt away and must be replaced from time to time.

Low voltage is usually used to avoid the decomposition of the cryolite.

EXTRACTION OF IRON

It's the most abundant metal in the earth's crust. The major ores from which Iron is extracted include the following;

Haematite, Fe_2O_3	Magnetite or lodestone, Fe_3O_4
Iron pyrites, FeS_2	Siderite or Spathic Iron ore, FeCO_3

Haematite, magnetite and limonite ores are in form of oxides and involve mainly 4 stages i.e. **concentration of the ore, reduction of the concentrated ore and purification of the extracted metal.**

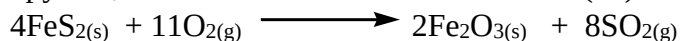
Iron pyrites and Siderite ores are not in form of the oxides hence involve mainly 4 stages i.e. **concentration of the ore, roasting of the concentrated ore, reduction of the roasted ore and purification of the extracted metal.**

Extraction process of iron from its ore:

Stage 1. Concentration of the ore by the froth floatation method: (for all the ores of iron)

Stage 2. Roasting of the concentrated ore: (for Iron pyrites and Siderite/Spathic iron ore)

For iron pyrites, FeS_2 it is roasted in air to form iron(III) oxide and sulphurdioxide.



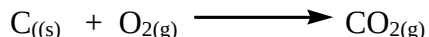
For siderite, FeCO_3 it is roasted in air to form iron(III) oxide and carbondioxide gas.



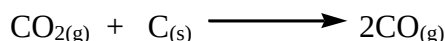
Stage 3. Smelting i.e Extraction of Iron by reduction of the oxide:

Either the concentrated ore (from haematite or magnetite) or the roasted ore (from iron pyrites or siderite), is mixed with coke and limestone and then introduced into a blast furnace where it is heated at high temperatures.

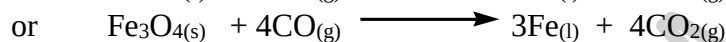
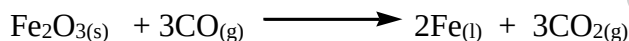
Near the base of the furnace, as the hot air passes through the mixture of roasted ore, coke and limestone, coke burns in the air blast to produce carbon dioxide.



As the carbon dioxide produced rises through the furnace, it is reduced by the excess hot coke to produce carbon monoxide.

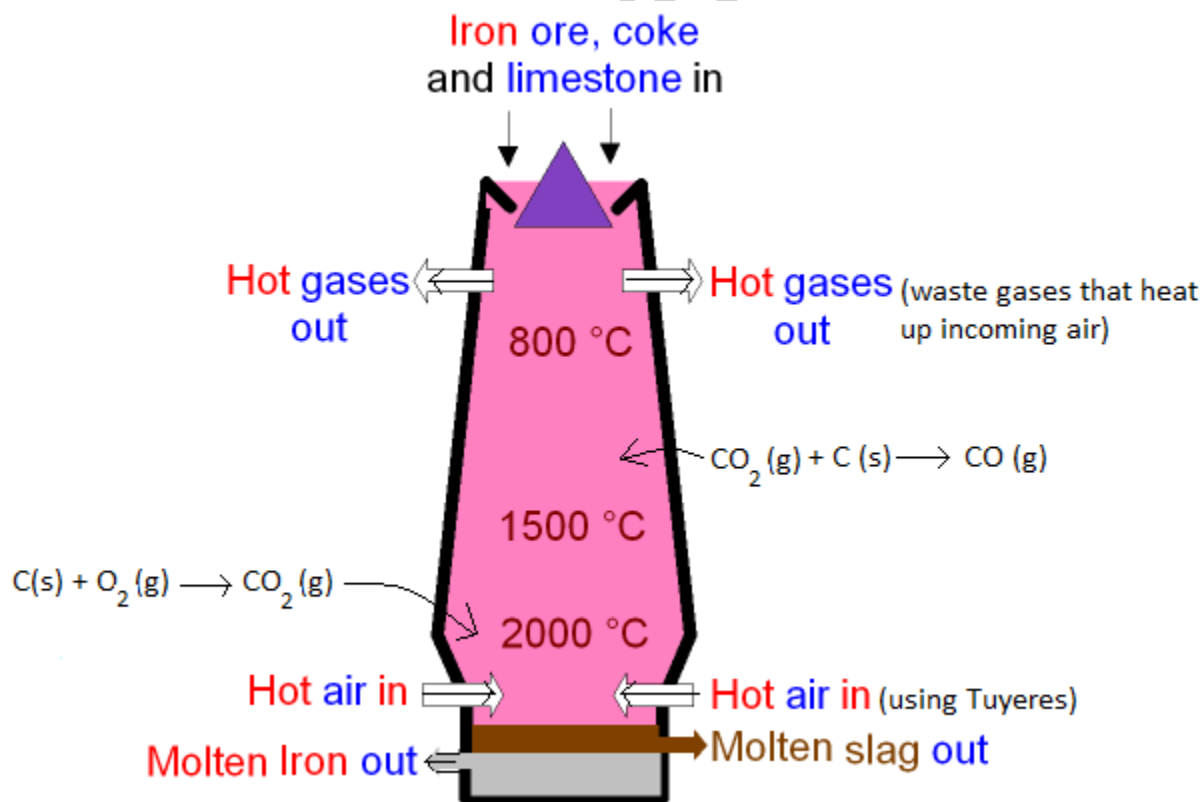


The reactions are highly exothermic and generate a lot of heat. The carbon monoxide ascends higher and reacts with the iron ore which is reduced to form molten iron as it is oxidised to carbondioxide.



Molten iron runs to the bottom of the furnace and is tapped off into moulds where it is solidified.

The moulds are called 'Pigs' and therefore this impure form of iron is called pig iron.



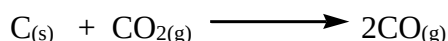
Roles of the limestone and coke:

Limestone which contains mainly calcium carbonate decomposes to form calcium oxide i.e. quick lime, (CaO) which reacts with the earthy impurities or silicon(IV) oxide and aluminium oxide to form slag (calcium silicate or calcium aluminate).

Thus, $\text{CaCO}_{3(s)} \longrightarrow \text{CaO}_{(s)} + \text{CO}_{2(g)}$

then $\text{CaO}_{(s)} + \text{SiO}_{2(s)} \longrightarrow \text{CaSiO}_{3(l)}$ or $\text{CaO}_{(s)} + \text{Al}_2\text{O}_{3(s)} \longrightarrow \text{Ca}(\text{AlO}_2)_{2(l)}$

Limestone also produces carbondioxide which reacts with more carbon to form carbon monoxide. Coke (carbon) when heated reacts with the carbon dioxide formed by limestone to form carbon monoxide which is also used to reduce the iron ore.



The molten slag floats on the top of molten iron as a separate layer at the bottom of the furnace. This is because the slag is less dense than iron. The slag protects the molten iron against any further oxidation by oxygen in the hot air in the blast furnace.

Waste gases include mainly carbon monoxide, carbon dioxide and nitrogen. The wastes gases escape from the top of the furnace. The large amount of carbon monoxide contained in the waste gases is burnt in special stoves and this reaction being exothermic preheats the air for the blast furnace which helps to reduce the energy costs of the process.

Stage 4. Purification of iron:

The carbon is usually removed by mixing the molten cast iron or pig iron with oxygen. The Bessemer process and the Open Hearth process are employed and the key principle is removal of impurities by oxidation with air blown into the molten iron.

Note that; the advantage of using the procedure described above to extract iron is because of the following factors:

- (i) Due to the cheap availability of coke.
- (ii) Coke has strong reducing properties and therefore it can easily reduce the metal ore.
- (iii) The wall of the blast furnace is made of bricks capable of withstanding the intense heat without melting or decomposing. Therefore the furnace can be used for a long period without repair being carried out.

Types of iron:

These are classified based on their percentage purity. The percentage purity also determines the strength and use of the iron. The types of iron are; pig iron, cast iron, wrought iron and steel.

(a) *Pig iron* is obtained directly from the extraction process. It is not pure as its found mixed with small traces of carbon.

(b) *Cast iron* is made when pig iron is remelted and silicon is added as an alloying element to improve its durability.

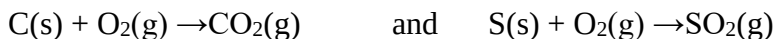
This is an impure iron which contains relatively high proportions of carbon (4%) and small proportions of other substances such as silicon, phosphorus and sulphur. Such impurities make cast iron to be hard, brittle and to have a lower melting point than pure iron. It cannot be welded and has little tensile strength.

The carbon is oxidized to carbon monoxide.

Cast iron can be used to make hot water pipes, Bunsen burner bases, cookers, railings and other purposes where little strain is imposed.

(c) *Wrought iron* is the purest form of iron (contains about 0.3% carbon) and is obtained from cast iron by heating it with iron(III) oxide in a furnace lined with limestone by a process known as puddling.

The oxygen from the iron(III) oxide oxidizes carbon and sulphur to their respective gaseous oxides and are removed as waste gases.



Phosphorus and silicon are oxidized to solid oxides.

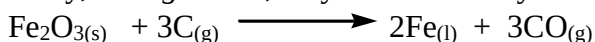


Calcium oxide (from decomposition of limestone) reacts with these solid oxides, forming calcium phosphate and calcium silicate, which float on the surface of the iron as slag.



The semi molten mass is then hammered and rolled so that the slag is squeezed out and a mass of almost pure iron remains.

Alternatively, *wrought iron*, may be obtained by direct reduction of haematite by carbon.



Wrought iron is very tough, malleable and ductile and is therefore used to make iron nails, iron sheets, ornamental work, horse shoes and agricultural implements. Wrought iron is sometimes referred to as low carbon steel.

(d) *Steel* is an alloy of mainly iron with carbon and other elements like manganese, chromium, silicon, cobalt and sometimes tungsten. The quality of steel depends on the amount of carbon present and this in turn determines its intended use.

Different types of steel are as shown below:

Steel	Composition	Uses
Mild steel	99% iron, 0.5% carbon	Car bodies, large structures
Hard steel	99% iron, 1% carbon	Cutting tools, razor blades
Manganese steel	87% iron, 13% manganese	Drill bits, springs
Stainless steel	74% iron, 18% chromium, 8% nickel	Cutlery, kitchen sinks, surgical instruments
Tungsten steel	95% iron, 5% tungsten	Edges of high-speed cutting tools

Steel is generally used in the construction of buildings, bridges, ships, car bodies, cutting and boring tools, crushing machines and stainless cutlery such as knives, forks etc.

Other General Uses of Iron

Making magnets, Iron sheets, Door and window frames, Rails, etc.

Question:

In the production of iron metal, Magnetite, Fe_3O_4 , is mixed with coke and lime stone and then heated in a blast furnace.

a) What is the purpose of adding:-

(i) limestone (ii) coke

b) Write three different equations for the reactions in which magnetite is converted to iron in the blast furnace.

c) The wall of the furnace is made of bricks. Explain why bricks are used.

d) (i) Name any three gases present in the “waste gases” emerging from top of the furnace.

(ii) Suggest how these gases could be made use of in the process.

e) (i) Name one other method that could be used to extract iron from its ore

(ii) Suggest a reason why the method you have named in e)(i) is not commonly used in the production of iron

f) State any three uses of iron

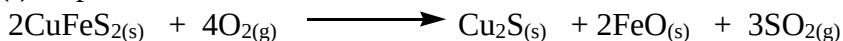
EXTRACTION OF COPPER

Some common ores include; copper pyrites (chalcopyrite) CuFeS_2 , malachite ($\text{CuCO}_3 \cdot \text{Cu(OH)}_2$), copper glance Cu_2S and cuprite Cu_2O .

The chief or major ore is copper pyrites.

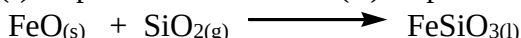
Stage 1. The ore is first concentrated by *froth floatation method*.

Stage 2. The concentrated ore is then **roasted in air**. Iron in the ore oxidises more readily than copper and is converted into Iron(II) oxide. The copper remains combined with sulphur as copper(I) sulphide, Cu_2S .

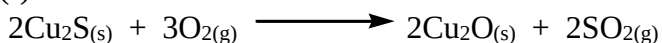


The sulphur dioxide gas formed can be used in manufacture of sulphuric acid as the non-metals present e.g. sulphur and phosphorous are eliminated as volatile oxides.

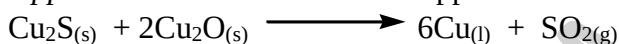
Stage 3. This is followed by **smelting**. The roasted ore, coke and silica (silicon dioxide) are placed in a water jacketed blast furnace when hot water is passed into the furnace together with silica (silicon dioxide), SiO_2 . All iron(II) sulphide is converted to iron(II) oxide. The iron(II) oxide is then removed by heating with a flux of silica in a furnace to form iron(II) silicate as slag which floats on the surface of the *fused matte* which is mixture containing mainly copper(I) sulphide with little iron(II) sulphide. The slag is poured off and removed.



As more air is blown, copper(I) sulphide reacts with oxygen and is partially converted to copper(I) oxide.



The copper(I) oxide then reacts with more copper(I) sulphide to produce impure copper called *blister copper* which contains 98.78% copper.



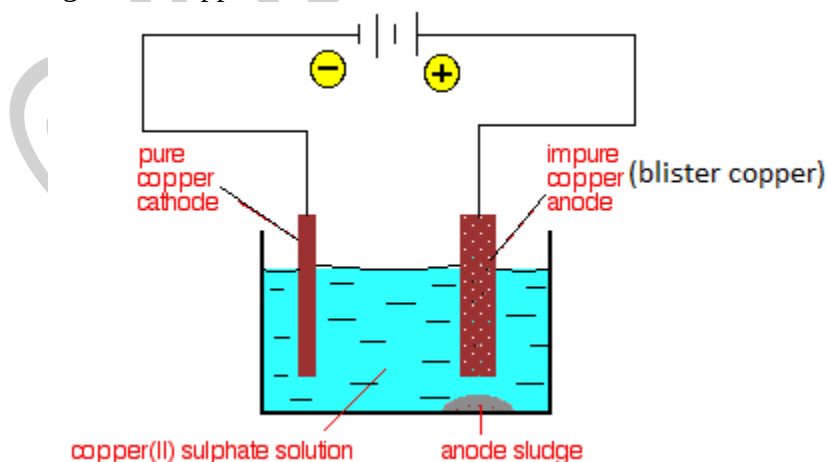
N:B (i) As long as iron(II) sulphide is present in the mixture, copper(I) oxide cannot be formed since copper has a higher affinity for sulphur than oxygen.

In molten state, iron(II) sulphide and copper(I) sulphide are miscible and the molten mixture is called **copper matte (fused matte)** which is converted into blister copper in a special furnace known as the **Bessemer converter**. More silica, SiO_2 , is added and an air blast is blown. This leads to formation of more slag.

(ii) In the reactions above, because of extremely high temperature employed, the products formed are in the **liquid** state. During solidification, sulphur dioxide gas escapes forming blisters on the surface of the metal thus the name blister copper.

Stage 4. Blister copper is then **purified or refined by process of electrolysis**:

An impure block of copper is made the anode and a thin pure strip of copper made the cathode. Electrolysis is then performed in a solution of copper(II) sulphate acidified with dilute sulphuric acid using a set of apparatus shown below.



During the electrolysis, pure copper is transferred from the impure anode to the cathode which is made of a strip of pure copper.

At the Anode; copper dissolves into the solution forming copper(II) ions as



At Cathode; copper is deposited as; $\text{Cu}_{(aq)}^{2+} + 2e \longrightarrow \text{Cu}_{(s)}$

Thus the cathode thickens with pure copper. Other impurities in the impure Anode which consists mainly of silver and gold are not discharged in the process but collect as sludge at the bottom of the cell, and they are subsequently recovered from it.

Question:

During refining of blister copper by electrolysis, a current of 5 amperes was passed through an electrolytic cell containing copper(II) sulphate solution for 3 hours 40 minutes and 50 seconds.

Calculate the mass of pure copper obtained. (Cu = 64, 1F = 96500 C)

Uses of Copper

- Used as an electrode in electrolysis and cells.
- For wiring electrical circuits as cables since it is a good conductor of electricity and is relatively cheap. The copper used for this purpose must be very pure since impurities increase electrical resistance.
- It is used for making ornaments like ear rings and pins, bangles e.t.c. being little attacked by air.
- It is used for making alloys like bronze (copper and tin) and brass (copper and zinc), copper coinage (copper and tin), German silver (copper, zinc and nickel)
- Used for making water pipes and boilers.
- Copper is used as a roofing material because it weathers to acquire a coating of green basic copper carbonate, $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2 \cdot n\text{H}_2\text{O}$, which gives a colourful touch to the building.

Question 1:

a) Outline the principles involved in metal extraction.

Solution

- Concentration of the ore
- Roasting of ore in air
- Reduction and Smelting of the ore
- Purification

b) Explain the following process as used in extraction of metals. Use examples where necessary.

i) Froth floatation ii) Roasting iii) Smelting

Question 2

a) Outline the various methods that can be used to extract metals. Name any two metals which can be extracted by each of those methods mentioned above.

b) Explain why the reduction process using coke is not employed in extraction of very reactive metals.

Solution:

This is because very reactive metals can easily combine with carbon to form carbides and hence not being able to reduce them. These metals also form very strong bonds in their oxides and therefore cannot easily be reduced by coke.

Question 3

a) Write the formulae of two ores of copper

b) Suggest one method by which the ores may be concentrated.

c) Outline, giving equations the process by which impure copper is obtained from the concentrated ore.

d) Name the method by which impure copper is purified.

e) Give any two large scale uses of copper.

EXTRACTION OF ZINC

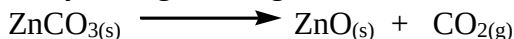
The chief major ores of Zinc are; Zinc blende, ZnS and Calamine, ZnCO_3

- The ore is first concentrated by **froth floatation** method.

- The concentrated ore is then **roasted in air** at about 900°C to form zinc oxide.

i.e for zinc blende $2\text{ZnS}_{(\text{s})} + 3\text{O}_{2(\text{g})} \longrightarrow 2\text{ZnO}_{(\text{s})} + 2\text{SO}_{2(\text{g})}$

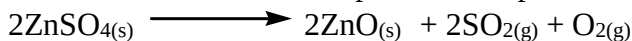
Alternatively during this stage, when calamine is heated, it decomposes to form zinc oxide.



During this process; some ore of the ZnS is also converted to zinc sulphate.



However above 900°C the zinc sulphate decomposes to form zinc oxide.



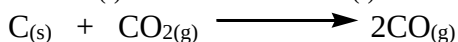
Overall equation is: $2\text{ZnS}_{(\text{s})} + 3\text{O}_{2(\text{g})} \longrightarrow 2\text{ZnO}_{(\text{s})} + 2\text{SO}_{2(\text{g})}$

The non-metals such carbon, sulphur and phosphorous are also eliminated as volatile oxides.

- This is followed by **extraction of zinc by reduction**.

The zinc oxide is crushed and then mixed with coke. Limestone is also added. The mixture is heated strongly in blast furnace, hot compressed air is also blown in. The coke reacts with carbondioxide from some heated coke or from limestone to form carbon monoxide which later reduces zinc oxide to zinc.

Thus $\text{CaCO}_{3(\text{s})} \longrightarrow \text{CaO}_{(\text{s})} + \text{CO}_{2(\text{g})}$

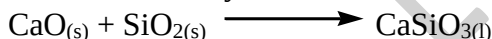


then $\text{CO}_{(\text{g})} + \text{ZnO}_{(\text{s})} \longrightarrow \text{Zn}_{(\text{g})} + \text{CO}_{2(\text{g})}$

Zinc is also formed by action of coke on zinc oxide being reduced to zinc as



Calcium oxide formed by limestone reacts with earthy materials forming slag;



The furnace used in this process is similar to that used in extraction of iron. Zinc has a boiling point of 1186K . The top of the furnace is maintained at about 1300K , so the zinc vapourises and distills off in absence of air (or in presence of carbon monoxide) in order not to be oxidised. The zinc vapour is cooled by a spray of molten lead and solidifies.

The crude form of zinc got is called **zinc spelter** (about 97% zinc mixed with lead, cadmium, etc) which can be further **purified** by fractional distillation or by electrolysis similar to blister copper.

Uses of Zinc

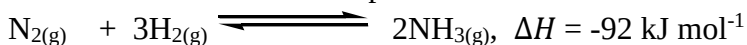
- It is used to galvanise iron
- It is used in making of alloys.

MANUFACTURE OF AMMONIA (The Haber Process)

The raw materials used for the manufacture of ammonia gas are namely Hydrogen and Nitrogen. Ammonia is obtained through direct combination of hydrogen gas with Nitrogen.

The reaction is slow at ordinary temperatures but its rate is increased by use of a catalyst. The catalyst used is finely divided iron powder or iron in form of wire gauze. It is made in this way so as to increase its surface area and allow easy flow of the gases over the catalyst. The catalyst is promoted by adding aluminium and potassium oxide. These promoters improve the stability of the catalyst and also increase its porosity thereby allowing a greater surface area to be used.

Equation for the reaction that takes place is:



Applying Le Chatelier's principle, the yields of ammonia at equilibrium can be increased by;

a) Increasing the pressure, as the forward reaction results in decrease in the number of molecules hence decrease in volume.

High pressure gives high yield of ammonia but the higher the pressure the greater the cost and maintenance hence becomes more expensive e.g. the pipes have to be thicker to withstand the pressure.

b) Decreasing the temperature as the forward reaction is exothermic.

A decrease in temperature or low temperature favour formation of more ammonia but will decrease the rate of reaction. Thus equilibrium conditions will take a longer time to achieve although a catalyst is used and thus being uneconomical.

Optimum conditions of temperature and pressure are therefore chosen. It does not decrease the yield of ammonia too much and yet maintain the reaction at a satisfactory rate. These lead to maximum yield of ammonia at minimum costs and they include:

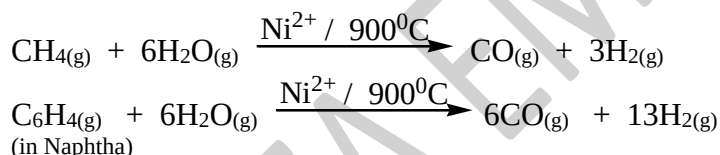
(i) Low temperature of 450°C.

(ii) High pressure of 200 to 350 atmospheres.

(iii) A catalyst is also employed to improve still further the rate at which equilibrium is attained satisfactorily at low temperatures. Low temperatures tend to slow the reaction.

Procedure:

- The nitrogen used in process is obtained by the fractional distillation of liquid air while hydrogen is obtained either from steam, natural gas or naphtha which is a mixture of hydrocarbons. Naphtha is one of the fractions obtained from the distillation of petroleum. Methane which is present in natural gas or naphtha is heated with steam at 900°C on a nickel catalyst. Thus



In each case synthesis gas is produced consisting of a mixture of carbon monoxide and hydrogen. Methane and Naphtha can also be partially oxidized by heating with a limited amount of oxygen.



More air is added to the reaction mixture which is then passed through a second reaction chamber where unreacted methane or naphtha is oxidized away otherwise they would affect the formation of ammonia.

- Compounds of oxygen and sulphur poison the catalyst in the haber process and have to be removed. During this process some carbon dioxide and water vapour is formed and so they have to be removed as they would also poison the catalyst. Water vapour is removed by passing the mixture through a drying agent such as concentrated sulphuric acid and then passed through a concentrated solution of potassium hydroxide under pressure to remove carbon dioxide gas. The carbon monoxide and remaining traces of carbon dioxide are removed by passing the mixture through ammoniacal copper(I) ethanoate solution.

- After obtaining hydrogen, the purified nitrogen and hydrogen mixture is compressed by pressure 200 between 350 atmospheres and passed over the catalyst at 450°C.

- During the reaction heat is evolved and is used to maintain the optimum temperature of the catalyst at 450°C.

- The hot gases leaving the converter pass through the heat exchange system and contains about 10 to 20% ammonia, the rest being Hydrogen and Nitrogen.
- The ammonia formed is removed by cooling the mixture to form the liquid (or absorbing the mixture in a specific amount of water). The ammonia liquefies when cooled and collects in storage vessels or since it is very soluble in water, it remains dissolved in it.
- The uncombined nitrogen and hydrogen pass through and re-cycled.

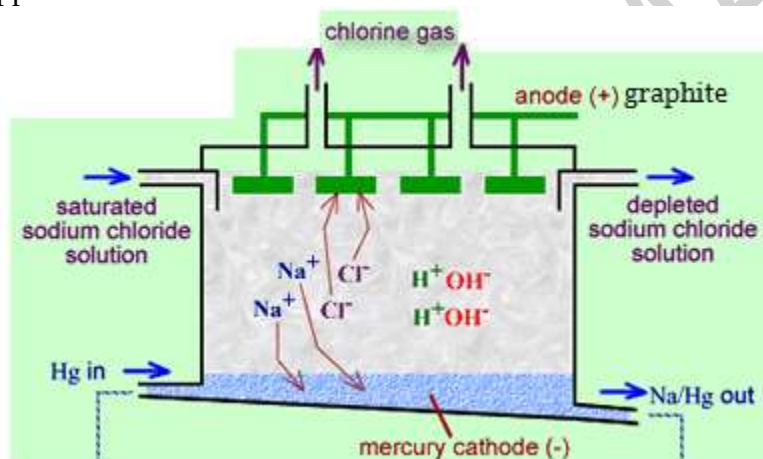
Uses of ammonia

Ammonia forms the basis of the nitrogen industry and has the following applications. It is used

- To manufacture fertilizers
- To manufacture nitric acid
- Used as a refrigerant (boiling point -33°C)
- Used in purification of water
- Used in manufacture of nylon as a synthetic fibre.

MANUFACTURE OF CHLORINE GAS AND SODIUM HYDROXIDE

(a) Chlorine gas and sodium hydroxide solution can be obtained industrially (manufactured) by electrolysis of a concentrated or saturated solution of sodium chloride (brine solution) as shown in the apparatus shown below.

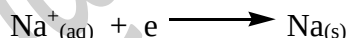


Concentrated or saturated sodium chloride solution (brine) is introduced into an electrolytic cell using the graphite / carbon anode and the mercury cathode as electrodes.

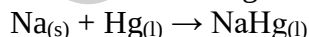
Chloride ions and hydroxide ions migrate to the anode. The chloride ions are more readily discharged in preference to hydroxide ions to liberate chlorine gas as shown by the equation:



Sodium is collected at the cathode as shown by the equation:



The sodium formed during the process dissolves in the flowing mercury forming an amalgam.



The amalgam is then dissolved in water from which sodium hydroxide is obtained.



The mercury does not dissolve in water and therefore is removed and recycled in the cell for use.

(b) Chlorine gas can alternatively be industrially obtained by electrolysis of molten sodium chloride using carbon (or titanium) anode and Iron / Steel cathode.

(Refer to extraction of sodium metal)

Uses of Chlorine include;

- used in making of bleaching agents
- in treatment of water (sterilizing)
- manufacture of organic chemicals e.g. CCl_4 , PVC

COMMERCIAL PREPARATION OF SODIUM CARBONATE (The Solvay process)

Soda ash is obtained naturally at Lake Magadi in Kenya.

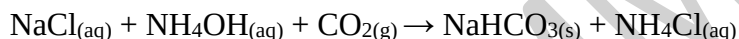
The raw materials for the Solvay process are *carbondioxide gas* generated from calcium carbonate (limestone), *sodium chloride* in form of brine and *ammonia* from the Haber process.

Procedure: Very concentrated brine (28% sodium chloride) is saturated with ammonia gas at the top of the Solvay tower to form ammoniacal brine (brine-ammonia mixture).

The ammoniacal brine then runs downwards the tower while carbon dioxide (formed from decomposition of calcium carbonate) is forced to rise up the tower from the base.

The tower is fitted with perforated mushroom shaped metal plates at intervals that delay the flow of ammoniacal brine and also offer surface area for the reaction.

The ammoniacal brine reacts with carbon dioxide to form sodium hydrogen carbonate as it precipitates at the lower part of the chamber since it is not very soluble in water. Precipitation is assisted by cooling the lowest third of the chamber.

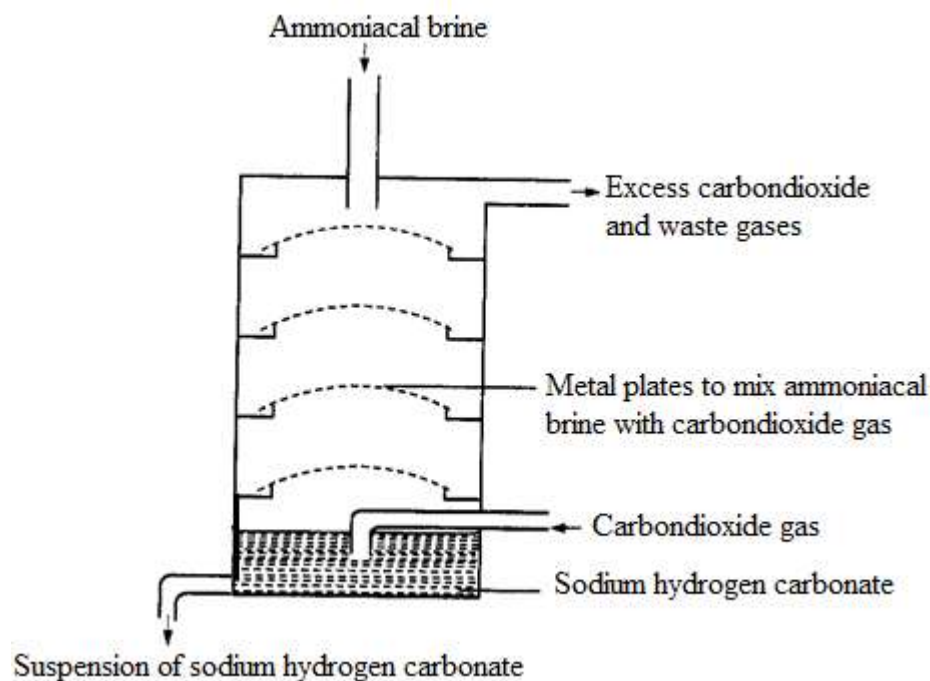


Sodium hydrogen carbonate is filtered from the white sludge at the base of the tower and washed to remove ammonium compounds.

The sodium hydrogen carbonate is then heated to form sodium carbonate.

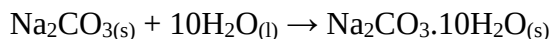


The anhydrous sodium carbonate formed has a wide market.



If crystalline form, **hydrated sodium carbonate (washing soda) i.e sodium carbonate decahydrate** is required, the anhydrous sodium carbonate (soda ash) is dissolved in hot water and crystallization takes place as the solution cools.

The crystals are removed and allowed to dry.



Note that:

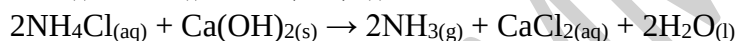
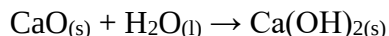
(a) Sodium carbonate decahydrate, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ are large translucent crystals. When the crystals are exposed to air, they lose mass and become coated with fine powder which makes it opaque. Each molecule of washing soda gives up to the atmosphere 9 molecules of water of crystallization to form sodium carbonate monohydrate.



Such loss of water of crystallization to the atmosphere is termed as efflorescence. The substance that loses its water of crystallization is known as an efflorescent substance.

(b) In the Solvay process, carbon dioxide is recycled for use.

Ammonia is recovered from the ammonium chloride by reacting ammonium chloride with calcium hydroxide, obtained by adding water to calcium oxide (from decomposition of calcium carbonate). Ammonia is recycled for use.



Therefore, the end products of Solvay process are calcium chloride and sodium carbonate.

Calcium chloride is used in extraction of sodium.

Uses of sodium carbonate:

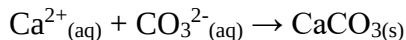
1. It is used in the manufacture of glass. Ordinary bottle glass is made by fusing (melting) together sodium carbonate, calcium carbonate, silicon dioxide (sand) and a little carbon (reducing agent). Broken glasses are added to assist fusion.

The resulting mixture of sodium and calcium silicates produces glass on cooling.

2. It is used in the manufacture of water glass that is used to preserve eggs, used in fire proofing and production of cement.

3. It is used in the manufacture of dry soap powders.

4. It is used for softening of water for domestic purpose. Calcium ions which are the principal cause of hardness in water are precipitated from water as calcium carbonate by the addition of sodium carbonate.



5. Used in the manufacture of sodium hydroxide (caustic soda).

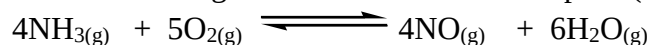
6. Used as a primary standard of acids in volumetric analysis.

MANUFACTURE OF NITRIC ACID (Ostwald's Process)

Nitric acid is manufactured by the oxidation of ammonia gas with the oxygen of the air in presence of a catalyst a metal gauze made of Platinum – Rhodium alloy.

Stage 1: Dry ammonia gas and dust free air are then heated and passed at a pressure of 98 to 99 atmospheres through a Platinum – Rhodium alloy gauze maintained at about 900°C. The

ammonia is converted to Nitrogen monoxide and water vapour (steam) on the surface of the catalyst.



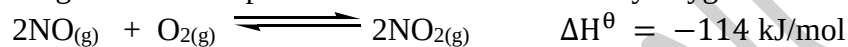
A lot of heat is given out which is sufficient to heat the catalyst. The optimum temperature of about 900°C is maintained by adjusting the rate of flow of the gases.

The maximum yield of nitrogen monoxide will be favoured by

- Low temperature since it is an exothermic process.
- Low pressure since the reaction proceeds with a decrease in volume of products.

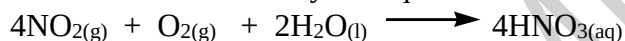
However, extremely low temperatures would make the reaction very slow and therefore uneconomical. Thus a slightly high pressure is used since there is slight decrease in volume. (i.e. 9 volumes to 10 volumes) and this ensures reasonable flow of the gases through the plant. After forming Nitrogen monoxide and water vapour, the hot gases leaving the converter are cooled.

Stage 2: The nitrogen monoxide produced is further oxidized by oxygen of the air for nitrogen dioxide.



The production of nitrogen dioxide is favoured by low temperature since the reaction is exothermic. Therefore nitrogen monoxide has to be cooled before being oxidized by more air to form nitrogen dioxide gas.

Stage 3: The nitrogen dioxide gas in presence of more air (oxygen) is then absorbed in warm water at about 350K. The presence of excess air avoids formation of nitrous acid and favours more formation of nitric acid by the equation



The acid produced contains about 60% of pure nitric acid. The acid can be made more concentrated using the 98% sulphuric acid solution as a dehydrating agent.

Concentrating the acid: Concentrated sulphuric acid is placed at the top of a tall tower packed with acid resistant rings. Nitric acid is heated by steam at the bottom of the tower. Nitric acid being more volatile distills off and its vapour is fed into a column to contact the concentrated sulphuric acid. The water is retained by concentrated sulphuric acid (a dehydrating agent) while pure nitric acid is obtained as a distillate.

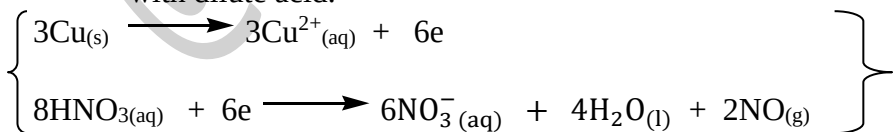
Uses of Nitric Acid:

- In manufacture of fertilizers such as ammonium nitrate and nitro – chalk (a mixture of ammonium nitrate and calcium carbonate)
- In manufacture of explosive being very volatile.
- In nitration of benzene and other organic compounds.

Some chemical properties of Nitric Acid:

(a) It is a strong oxidizing agent: Nitric acid oxidizes some metals to their corresponding metal ions e.g.

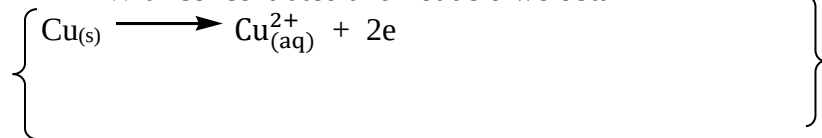
- with dilute acid.

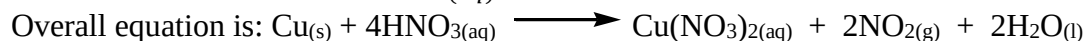
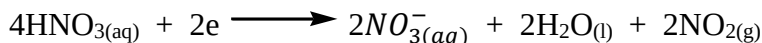


Overall equation is; $3\text{Cu}_{(s)} + 8\text{HNO}_{3(aq)} \longrightarrow 3\text{Cu}(\text{NO}_3)_{2(aq)} + 2\text{NO}_{(g)} + 4\text{H}_2\text{O}_{(l)}$

However with magnesium metal, magnesium nitrate and hydrogen gas are formed.

- With concentrated and hot acid we obtain





However, nitric acid has no effect on some metals such as gold and platinum. Similarly it has no effect on other metals like aluminium, iron, chromium etc.

This is because of formation of impervious oxide film formed on surface of these metals, a phenomenon known as passivity i.e. they are passive to nitric acid.

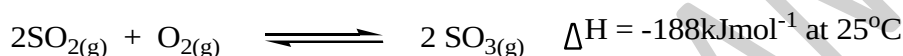
Other non-metallic elements are oxidized by nitric acid



(b) It undergoes other common reactions of dilute acids with bases, carbonates, hydrogencarbonates.

MANUFACTURE OF SULPHURIC ACID (The Contact Process)

Manufacture of sulphuric acid in which the raw materials sulphurdioxide and oxygen are passed over vanadium(V) oxide catalyst at optimum pressure of between 1 to 3 atmospheres at temperature of about 450°C.



Here the yield of SO_3 can be increased by;

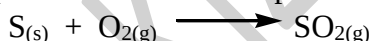
- Use of low temperature since the forward reaction is exothermic.
- Use of high pressure since the forward reaction occurs with decrease in volume.
- Increasing concentration of either SO_2 or O_2 but excess O_2 is preferred as it's cheaply obtained from air.
- Decrease in concentration of SO_3 by constantly removing as soon as it's formed.

Note that: In the contact process, care should always be taken so as to keep the air supply down to a minimum. This is because the presence of inert nitrogen gas will reduce the yield. For any equilibrium reaction, which occurs with decrease in number of moles, addition of inert gas favours the backward reaction.

Stages involved in manufacture of sulphuric acid:

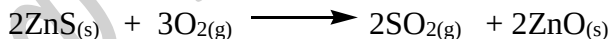
Stage 1: The first stage in manufacture of sulphuric acid involves formation of sulphur dioxide gas. This is obtained by burning sulphur or roasting sulphide ores in air which act as the raw materials in production of sulphuric acid. Thus roasting:

(a) sulphur in air to form sulphur dioxide.

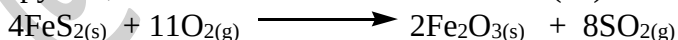


(b) concentrated sulphide ores:

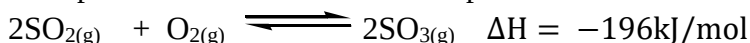
(i) Zinc blende, ZnS is roasted in air to form zinc oxide and sulphurdioxide.



(ii) Iron pyrites, FeS_2 is roasted in air to form iron(III) oxide and sulphurdioxide.



Stage 2: The sulphur dioxide formed is then mixed with excess air and thoroughly purified. This prevents poisoning of the catalyst by dust and other impurities. The mixture is then treated to optimum conditions which involve use of a catalyst Vanadium(V) oxide, V_2O_5 by contact process, at temperature range of (400 to 450°C) and a pressure slightly greater than atmospheric pressure. Sulphur dioxide is oxidized to sulphur trioxide.



Note that; vanadium pentoxide has the advantage of not being poisoned easily by impurities and it is cheaper to be obtained. The mixture of purified air and sulphurdioxide is cooled from 600°C

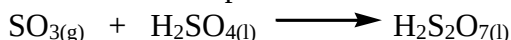
to about 430°C by passing it through a heat exchanger. The heat removed is used to heat the limited mixture of sulphur dioxide and air.

Platinum which has been previously used as catalyst is not employed because it is expensive and highly susceptible to poisoning. Iron(III) oxide which can also function as a catalyst has the disadvantage of requiring the highest temperature moreover with lower yield. The catalyst used (V_2O_5) assists only in achieving the equilibrium in a much shorter time but does not increase the yield of sulphur trioxide.

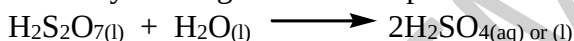
It is made in cylindrical shape as this increases the surface area of the catalyst and allows easy flow of the gases.

Since the reaction is exothermic and proceeds with a decrease in volume the yield of sulphur trioxide is increased by low temperature and high pressure. However extremely low temperatures would make the reaction slow thus the equilibrium would not be achieved faster. Very high pressure may cause leakage and lead to maintenance of equipment very expensive. Therefore optimum conditions which involve use of a catalyst Vanadium(V) oxide, a temperature range of (400 to 450°C) and a pressure slightly greater than atmospheric are employed or applied. These lead to maximum yield of sulphur trioxide hence maximum production of sulphuric acid.

Stage 3: The sulphur trioxide obtained above is cooled and absorbed in a little appropriate amount concentrated sulphuric acid to form fuming sulphuric acid or oleum.



Stage 4: After cooling the oleum it's then stored. Sulphuric acid of any required concentration is then obtained by diluting oleum with specific amount of water.



Noted that; sulphur trioxide (SO_3) is not dissolved directly in water to produce sulphuric acid. This is because the reaction is highly exothermic. Therefore as such highly corrosive misty fumes of sulphuric acid will be formed which would combine with water vapour in the atmosphere to form tiny droplets which would condense down hence being dangerous to the environment and people working in the industry. However when sulphur trioxide is absorbed in concentrated sulphuric acid it does not evaporate easily.

When diluting sulphuric acid, water is never added to the acid. Instead it must be the acid which should be added to water placed in a container. This is because if the water is added, since the reaction is exothermic, the little amount of water added (acid being in greater amount) would produce a lot of heat which would make the acid to spill. However if a little amount of acid is diluted more easily hence a lot of heat is not suddenly generated.

The emerging gases are recycled to remove any sulphur dioxide and sulphur trioxide present.

Some chemical properties of sulphuric acid:

a) Concentrated sulphuric acid is a strong oxidizing agent. e.g. it is able to:

- Oxidize sulphur to sulphur dioxide
 $S_{(s)} + 2H_2SO_{4(l)} \longrightarrow 2H_2O(l) + 3SO_{2(g)}$
- Oxidize copper to copper (II) ions
 $Cu_{(s)} + 2H_2SO_{4(l)} \longrightarrow CuSO_{4(aq)} + SO_{2(g)} + 2H_2O(l)$
- Oxidises hydrogen iodide and hydrogen bromide to iodine and bromine respectively.
- etc

b) Concentrated sulphuric acid is a strong dehydrating agent able to dehydrate copper(II) sulphate pentahydrate (blue) to white anhydrous copper(II) sulphate, etc.

c) Undergoes common reactions of dilute acids with metals, bases, carbonates, hydrogencarbonates.

Uses of sulphuric acid:

- Used in making of fertilizers such as ammonium sulphate.
- Used as a dehydrating agent e.g. in drying of gases.
- Used as a laboratory reagent
- In preparation of alkenes

Some industries produce a large amount of SO₂ and CO₂ in the air. These gases sometimes mix in the water vapour in the atmosphere or in rain as it falls to the ground. Thus acidic droplets or rain main result which corrodes iron roof of some structures or polluting the environment.

FERTILIZERS

A **fertilizer** is an artificial or natural substance containing essential elements that are added to soil in order to make it more productive for the healthy growth of plants. Fertilizers provide essential nutrients elements for plant growth. These nutrients include nitrogen, phosphorous and potassium. Intensive cropping of agricultural land means that nitrogen removed from the soil can be replaced.

Examples of natural fertilizers include manure and compost.

Examples of artificial fertilizers include Ammonium nitrate (Nitram), Nitro chalk (ammonium nitrate coated with chalk), NPK (Nitrogen phosphorous and potassium), CAN (Calcium, Aluminium, Nitrogen), Ammonium sulphate.

Artificial or Inorganic Fertilizers

These are synthetic fertilizers that contain a high percentage of essential elements i.e. nitrogen, phosphorus and potassium; the fertilizers should be cheap and soluble in water.

The examples of such fertilizers include: Ammonium nitrate (Nitram), Nitro chalk (ammonium nitrate coated with chalk), NPK (Nitrogen phosphorous and potassium), CAN (Calcium, Aluminium, Nitrogen), Ammonium sulphate, Ammonium phosphate, Potassium nitrate, Liquid ammonia and Potassium chloride.

Note that:

- (a) Liquid ammonia is very soluble; it can easily be washed away by rain water and can easily evaporate off the soil.
- (b) A composite/balanced fertilizer contains a combination of elements in certain correct proportions e.g. nitrogen, phosphorus and potassium which is commonly labeled NPK.
- (c) Excessive use of ammonium fertilizers renders the soil acidic. It is therefore necessary to add quick lime (calcium oxide) to neutralize the acidity. The added lime also makes the soil more porous for proper air and water absorption.
- (d) Nitrogen can also be incorporated into the soil by nitrogen fixing bacteria.

Manufacture of Fertilizers:

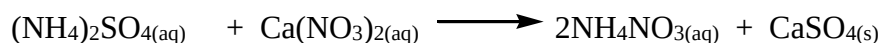
(a) Ammonium Nitrate.

On the large scale, ammonium nitrate is made by reacting or passing ammonia gas through concentrated nitric acid.

$$\text{NH}_{3(g)} + \text{HNO}_{3(aq)} \longrightarrow \text{NH}_4\text{NO}_{3(aq)}$$

Ammonium nitrate is left to crystallise out.

Alternatively ammonium nitrate can be obtained by reacting concentrated solution of ammonium sulphate and calcium nitrate.

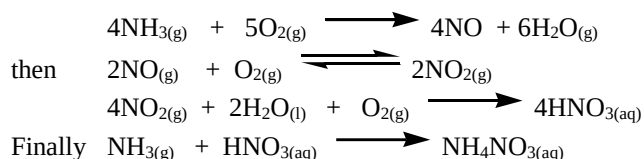


The sparingly soluble calcium sulphate is filtered off and the ammonium nitrate is crystallized from the solution.

Question

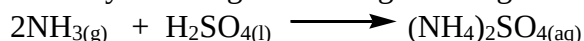
Using only ammonia, air and water as the only reagent explain how ammonium nitrate can be obtained.

Solution



(b) Ammonium Sulphate.

It is obtained by bubbling ammonia gas through a concentrated solution of sulphuric acid.

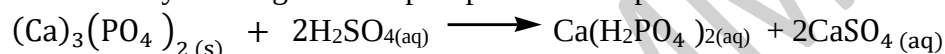


(c) Fertilizers of phosphorous:

All naturally occurring fertilizers of phosphorous are the phosphates(V) the most common of these being rock phosphate (calcium phosphate) $(\text{Ca})_3(\text{PO}_4)_2$ the source of phosphorous and other phosphates. Rock phosphate is the mostly consumed by the fertilizer industry in the manufacture of super phosphate of lime, triple super phosphate and 'nitrophos' (a combined phosphatic and nitrogenous fertilizer)

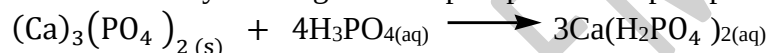
Super phosphate:

It is made by reacting calcium phosphate and sulphuric acid.



Triple super phosphate:

It is obtained by reacting calcium phosphate with phosphoric acid



Nitro phos:

Obtained by reacting calcium phosphate and nitric acid



Advantages of artificial fertilizers include;

- They do not harbour pests and diseases.
- They are available for use whenever needed.
- They are effective when used in small amounts.
- Different fertilizers for different crops and soil types are usually available.
- They normally give good yield.

Disadvantages of artificial fertilizers include;

- They are expensive in terms of production costs.
- They do not give desired results if the instructions for application are not strictly followed.
- They may exhaust soil fertility if used for a long time.
- The crop/yield is of inferior quality to one produced using organic fertilizers.
- When they are washed down into the lakes and rivers cause pollution. i.e algae bloom.
- Excess use of fertilizers may lead to the soil being acidic.

Natural or Organic Fertilizers

These are fertilizers from decomposed animal and plant waste matter such as cow dung, chicken droppings, dead animals, dead leaves, plant branches, coffee husks etc.

Examples of natural fertilizers include manure and compost.

Advantages of organic fertilizers include;

- They are cheap to make since they are prepared by the farmer.
- They have no elaborate instruction for application.
- They provide a suitable habitat for nitrifying bacteria.
- They last for a long time in the soil thus improving soil texture.
- They hold water thus keeping the soil moist even in dry season.

Disadvantages of organic fertilizers include;

- They cannot be used selectively i.e. on soil which lacks a particular nutrient.
- They harbor pests and diseases which attack the crop.
- They have to be applied in bulk for effectiveness.
- They act slowly and are not ready for use any time since they have to first be dried.
- When they are washed down into the lakes and rivers cause pollution. i.e algae bloom.
- Excess use of fertilizers may lead to the soil being acidic.

MANUFACTURE OF BIO GAS

Bio gas is a mixture of gases with a high methane content which is produced by microbial fermentation of organic wastes in absence of oxygen gas and can be used as a source of fuel. However, *bio mass* is renewable energy source of living or recently living plant and animal materials that can be used as a source of fuel.

Biogas is formed when bacteria acts on animal and vegetable wastes.

Main Composition of bio gas is methane gas (CH_4). Other gases present include; ammonia, hydrogen sulphide, carbon dioxide, carbon monoxide, steam, etc.

Simple Biogas Generator:

It consists of a container in which animal and vegetable wastes are mixed with a correct (limited) amount of water and then covered to prevent atmospheric oxidation (to exclude aerial oxidation). A temperature of about 25°C to 30°C is maintained. The bacteria present in the wastes break down (decompose) the waste to form biogas.

The quality of the biogas depends on the type of waste. For instance, mixture of human excreta or cow dung and bean stalks produces high quality biogas.

Uses: source of fuel for cooking and heating as well as lighting.

Advantages related to biogas production include:

- (a) Biogas is easy and cheap to produce since it is produced from waste matter as a byproduct.
- (b) Forests and wild life would be saved if institutions and rural communities turned to using biogas instead of wood and charcoal for their fuel needs.
- (c) Sewage materials and other biological wastes can be converted to biogas.
- (d) Solid wastes from biogas can be used as fertilizers because of its high nitrogen content.
- (e) Produces less pollutant gases. i.e. It's a cleaner fuel

Disadvantages related to biogas production include:

- (a) Release of carbon dioxide which pollutes the environment and leads to global warming.
- (b) Production of sulphur dioxide and carbondioxide gas which are pollutant gases and responsible for acid rain.

Review Questions:

1. (a) What is meant by biogas and how is biogas produced.
(b) Outline the advantages related to biogas production.
2. (a) State the difference between bio mass and bio gas.

- (b) (i) Name the main component of bio gas.
(ii) Write an equation for complete combustion of the gas named in (b) (i) above.
(c) State two advantages of using bio gas fuel instead of charcoal as the major source of fuel in homes.

TREATMENT OF SEWAGE

Sewage is faecal and waste water matter from toilets, bathrooms, factories, and kitchens in communities. Sewage is normally carried away by underground pipes called sewers into prepared areas called lagoons for treatment.

Treatment of sewage involves the following;

- (a) Removal of suspended materials (solids).
- (b) Break down of organic and inorganic matter by aerobic bacteria. The bacteria uses the oxygen present in the sewage water to oxidize the pollutants into harmless materials.
- (i) Oxygen is increased by blowing air into the sewage tanks.
- (ii) Suitable pH and temperature (25 to 35°C) and good aeration are important conditions for decomposition and oxidation of the pollutants.
- (c) Finally the solid (sludge) and liquid (effluent) are separated.

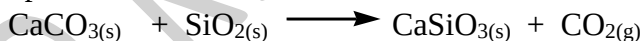
The effluent is allowed to join wetlands for natural filtration before flowing into the rivers and / or lakes.

The sludge may be used as follows;

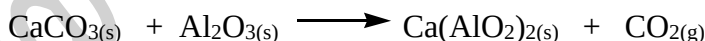
- (a) For road surfacing as it can set into hard solid.
- (b) As a fertilizer – it contains phosphorous and nitrogen.
- (c) As a raw material for production of biogas
- (d) To fill quarries for land reclamation.

MANUFACTURE OF CEMENT $\text{Ca}(\text{AlO}_2)_2 / \text{Al}_2(\text{SiO}_2)_3$

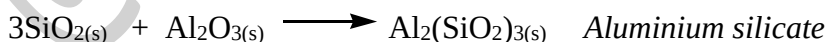
Portland cement is an important building binder material. A cement is a substance used for construction that sets, hardens and adheres to other materials, binding them together. It is manufactured by mixing limestone and clay. Limestone mainly contains calcium carbonate while clay mainly contains silicon(IV) oxide, aluminium oxide and water. Thus clay is basically aluminosilicate hydrate. The mixture of limestone and clay is well crushed and then roasted / heated in a rotary kiln at about 1500°C. This produces a grey mass called clinker which is ground to a fine powder and sold as cement. The reactions that takes place during roasting process are



Calcium silicate



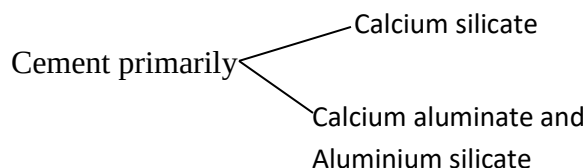
Calcium aluminate



As the equations show, cement contains silicates and aluminates.

Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) is added because of its absorption of water.

Other blends include; Bauxite, laterites, heamatite



By products are:

Carbon dioxide and sulphur dioxide which are pollutants, also included are dust and smoke.

Concrete is made when cement is mixed with sand and water in right proportions. It is allowed to set for some time, as a complicated hydrolysis reaction takes place and an interlocking silicate network emerges which causes the mass to harden forming a hard stone that is resistant to corrosion.

Uses of cement are mainly in

- i) construction and joining of bricks (floor and wall plastering)
- ii) building of dams, bridges and roads.